

OpenClaw Core Concepts

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2. Course Content

This chapter systematically explains the seven core concepts of OpenClaw:

1. **Chat Channels** - The platform where AI communicates with users
 2. **Agent** - The core brain of the AI agent
 3. **Sessions & Memory** - Context management capability
 4. **Tools** - The hands and feet that execute specific operations
 5. **Skills** - Encapsulated functional modules
 6. **Models** - The engine that drives AI thinking
 7. **Gateway** - The nervous center of the system
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3. Core Concepts

3.1 Chat Channels

What are chat channels?

Chat channels are the communication platforms where users interact with AI Agents. Think of them as the "desk" or "workplace" of an AI employee.

Supported channel types:

Channel Type	Representative Platforms
Instant Messaging	Slack, Discord, Microsoft Teams
Social Apps	Telegram, WhatsApp, Signal
Web Interface	WebChat, REST API

Core features of channels:

- **Plug and play** - Enable the desired channel without redeploying

- **Many-to-many binding** - One channel can connect to multiple Agents, and one Agent can also serve multiple channels
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3.2 Agent

Agent is the core decision-making unit of OpenClaw. Think of it as your "virtual employee" or "brain". Each Agent has its own independent:

- **Persona prompt** - Defines the AI's behavioral style and role
- **Model selection** - Determines which LLM to use (such as GPT-4, Claude, etc.)
- **Channel binding** - Specifies which platforms to work on

Key features:

- **Multi-agent parallelism** - A single OpenClaw instance can run multiple Agents
 - **Role isolation** - Different Agents do not interfere with each other and are configured independently
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3.3 Sessions and Memory

Why is memory needed?

The session and memory system allows the AI to remember context and provide a coherent conversation experience.

OpenClaw's three-tier memory architecture:

Short-term Memory

- Storage format: daily log file (`memory/yyyy-mm-dd.md`)
- Retention period: conversations from the last 1-2 days
- Purpose: context for handling the current task

Near-term Memory

- Storage location: `sessions/` directory
- Management method: sliding window compression, keeping complete session archives
- Purpose: context tracking for multi-turn conversations

Long-term Memory

- Storage content: user preferences, important decisions, historical experience
 - Loading method: automatically load `memory.md` at the start of each conversation
 - Purpose: achieving personalized intelligence and continuous learning
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3.4 Tools

What are tools?

Tools are the "hands, feet, and senses" through which an Agent interacts with the external world, allowing the AI to perform real actions. If the LLM is the brain, then tools are the limbs attached to the brain.

OpenClaw built-in tools:

Tool Type	Description	Use Cases
File system	Read/write files, create directories, search content	Document processing, data organization
Network requests	HTTP API calls, Webhook triggers	Third-party service integration
Shell commands	Execute system commands, run scripts	Automated operations, batch processing
Scheduled tasks	Cron scheduling, periodic execution	Scheduled backups, daily report delivery

Tool execution flow:

User instruction → Agent understands intent → selects the appropriate tool → executes the action → returns the result

3.5 Skills

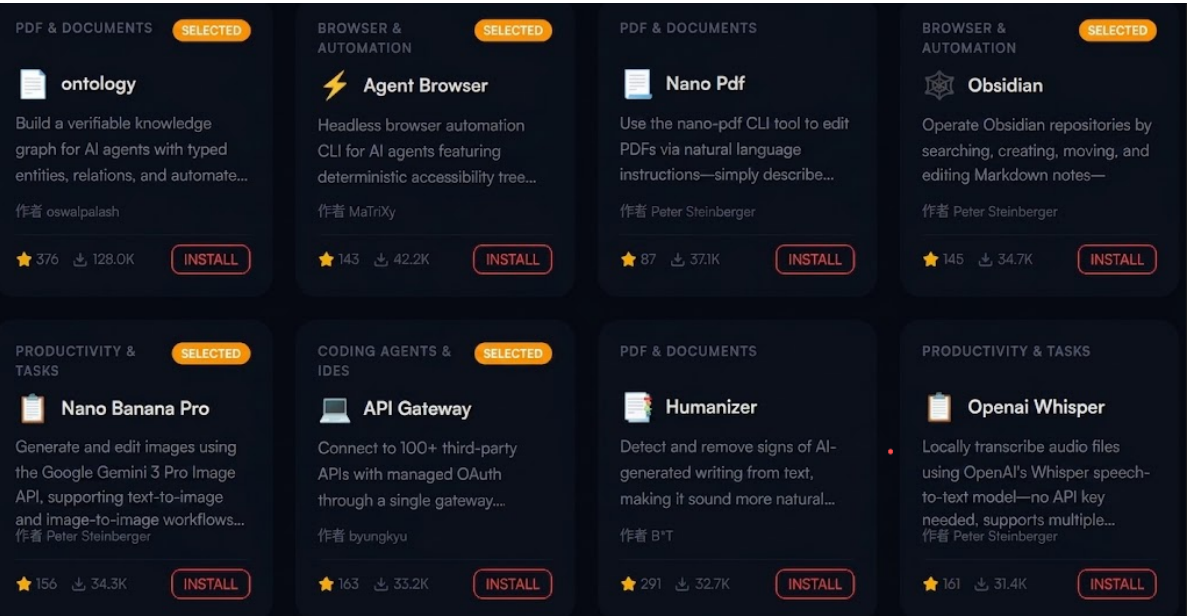
Skills vs. tools: what's the difference?

- **Tools** are basic capabilities, such as "open a file" or "click the mouse"
- **Skills** are encapsulated business logic, such as "grasp an object with the robotic arm" or "navigate to a certain location"

An intuitive analogy for skills:

If an Agent is compared to a robot, then:

- **Agent** = the robot's brain (responsible for thinking and decision-making)
- **Tools** = the robot's hands and feet (responsible for performing basic actions)
- **Skills** = the robot's training manual (tells the robot how to complete a specific job)



Core features of skills:

- **Atomicity** - A skill does only one thing (such as "grasp an object")
- **No decision-making** - Skills execute passively and do not judge "whether to do it"

- **Standardized** - Follow OpenClaw conventions and can be reused across scenarios
- **Pluggable** - Can be installed/uninstalled at any time

Skill sources:

- **ClawHub** - A community-maintained skill marketplace
- **Custom skills** - Written by developers themselves (in Markdown or TypeScript)

3.6 Models

What are connected models?

Models refer to the large language models (LLMs) that drive Agent thinking; they are the "intelligence engine" of AI. OpenClaw itself does not train models, but acts as a "middle layer" that connects to various LLM providers.

The role of models:

- **Understanding intent** - Parses the natural language of user input
- **Task planning** - Breaks complex goals down into executable steps
- **Tool selection** - Decides which tool or skill to call
- **Self-correction** - Adjusts strategies based on execution results

Supported model providers:

Provider	Representative Models	Features
OpenAI	GPT-4o, GPT-4 Turbo	Strong overall capability, mature ecosystem
Anthropic	Claude 3.5, Claude Sonnet	Excellent long-text processing, high security
Google	Gemini Pro, Gemini Ultra	Strong multimodal capability
Mistral	Mistral Large, Mixtral	Open-source models, high cost-effectiveness
Ollama	Llama 3, Qwen, etc.	Local deployment, privacy protection
Others	Any OpenAI-compatible interface	Custom model integration

3.7 Gateway Architecture

What is Gateway?

Gateway is the "nervous center" and "commander-in-chief" of OpenClaw, and a long-running background service. It centrally manages the connections of all communication channels and coordinates communication among Agents, clients, and nodes.

Core responsibilities of Gateway:

Message routing

- Receive messages from 50+ channels (WhatsApp, Telegram, Slack, etc.)
- Verify user identity and permissions
- Distribute to the corresponding Agent based on the session ID

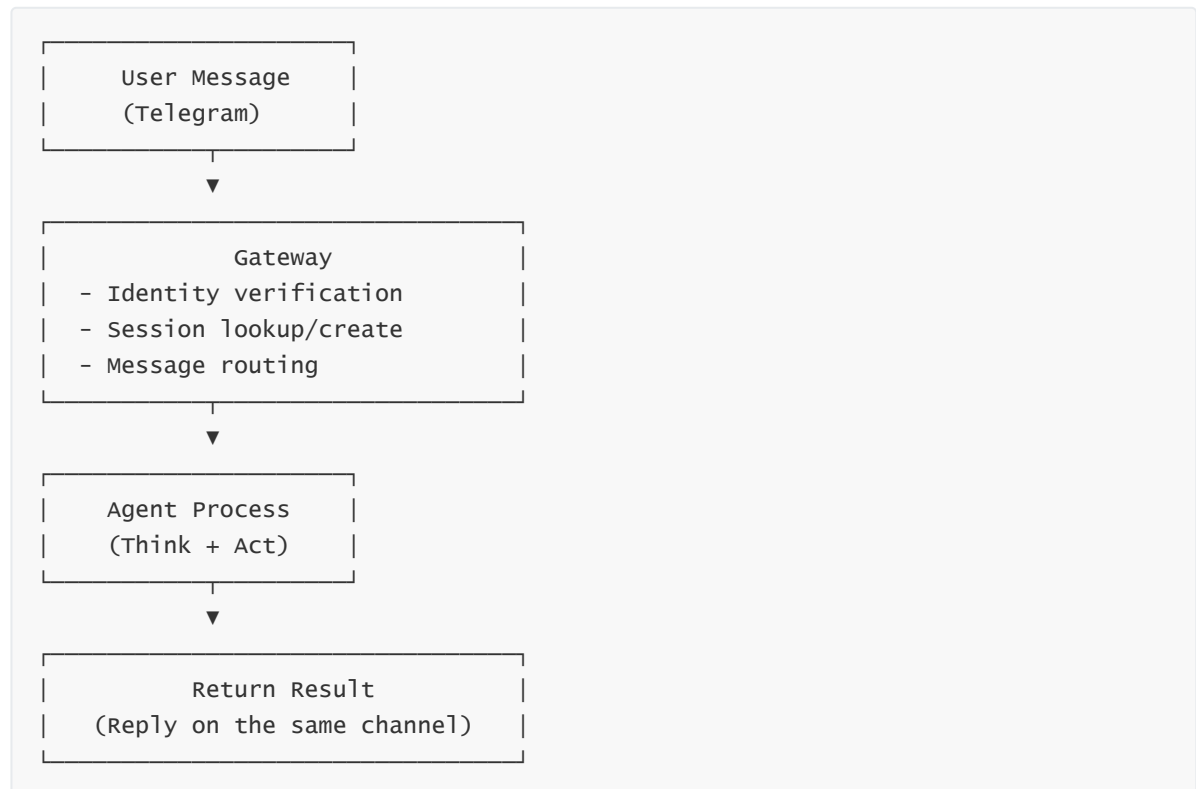
Session management

- Maintain the state of all sessions
- Support "explicitly parallel, serial by default" processing
- Heartbeat detection and exception retry

Task scheduling

- Scheduled task triggering (Cron)
- Load balancing and failover
- Performance monitoring and logging

Gateway workflow:



4. Core Concept Relationship Diagram

One-sentence summary of each concept's role:

Concept	Intuitive Analogy	Core Role
Chat Channel	Office/Desk	User interaction entry point
Agent	Virtual employee/Brain	Decision-making and planning
Sessions & Memory	Work notes/Experience	Context management
Tools	Hands, feet, and senses	Basic execution capability
Skills	Training manual	Encapsulate business logic and processes
Models	Intelligence engine	Thinking and reasoning
Gateway	Nervous center	Scheduling and coordination

5. Chapter Summary

- OpenClaw is a local-first AI Agent operating system, not a simple chatbot
- The role positioning and interrelationships of the seven core concepts
- The hierarchical architecture design of channels, Agents, skills, and tools
- Supported model providers and selection strategies
- The key role of Gateway as the system hub